

Observational studies

An observational study records conditions without assigning treatments.

Students choose whether to attend tutoring. We compare their exam scores. What kind of study is this?

Experiments

An experiment assigns treatments and measures outcomes. A randomized experiment uses chance to assign the treatments.

A teacher randomly assigns two practice formats. What makes this an experiment?

Experimental units

An experimental unit is an individual or object that receives a treatment.

Thirty printers are each assigned one of two paper types. What are the experimental units?

Explanatory variables

The explanatory variable, also called the independent variable, is the proposed influence on an outcome.

A study compares two paper types for their effect on printer jams. What is the explanatory variable?

Response variables

The response variable, also called the dependent variable, is the measured outcome.

How could we measure printer performance in the paper-type study?

Treatments

Treatments are the experimental conditions assigned to units.

A printer study uses standard, recycled, and coated paper. What are the treatments?

Lurking variables

A lurking variable is an unaccounted-for influence on a study.

Students who choose tutoring earn higher scores. What other influence might explain the difference?

Confounding

Confounding prevents us from separating the effects of different influences.

One practice format is taught only by Teacher A, the other only by Teacher B. What is confounded?

Random assignment

Random assignment uses chance to allocate treatments.

How could we assign 40 volunteers to two practice formats?

Chance imbalance

Forty volunteers are randomly assigned to two practice formats.

Must the two groups begin with exactly equal prior knowledge?

Random sampling or assignment?

A researcher randomly selects 100 students, then lets each choose a practice format.

Which part is random sampling, and is there random assignment?

Control groups

A control group provides a comparison condition.

What would be a useful control group for a study of a new practice app?

Placebos

A placebo resembles a treatment but lacks its active component.

Why might an inactive version of a treatment be useful in a comparison?

Placebo effects

A placebo effect is a response associated with expectations of treatment.

How could a participant's expectations change a self-reported outcome?

Blinding

Blinding, also called masking, withholds treatment identities from specified people.

Why might a grader receive assignments without knowing each student's practice group?

Double blinding

In a double-blind experiment, neither participants nor the people measuring outcomes know the treatment identities.

Could students be blinded to whether their practice uses paper or an app?

Replication

Replication applies each treatment to multiple independent experimental units.

Why would one printer per paper type be weak evidence?

Causal conclusions

Students who voluntarily use a new study app score higher than students who do not.

Does this establish that the app caused the difference?

A study design

Forty volunteers will compare two versions of a practice app using the same final quiz.

How would you design a fair experiment?

When assignment is impossible

A study compares performance across naturally occurring academic-year groups. Researchers cannot randomly assign a student's year of study.

How does this limit the study's ability to establish causation?

Treatment order

Every volunteer tries two practice formats.
Randomly choose which format each person uses first.

Why might using the same format first for everyone confound the comparison?

The parts of an experiment

Synthetic study: randomly select 40 campus printers, assign standard or recycled paper, and count jams per 1,000 pages.

What are the population, sample, experimental units, explanatory variable, response variable, and treatments?

Participation and consent

A researcher tells students that refusing a study will lower their course grades.

What is ethically wrong with this recruitment?

Informed consent

Informed consent is voluntary agreement after understanding the study, its risks, costs, and potential benefits.

What information should a participant receive before agreeing?

Institutional Review Board

An Institutional Review Board (IRB) reviews proposed human-subject research and protections for participants.

Why should review occur before recruitment and data collection?

Risk and benefit

Ethical research minimizes risks and weighs them against anticipated benefits.

How could a driving simulator reduce the risks of studying distraction?

Privacy

A public spreadsheet lists participant names beside individual quiz scores.

How should these data be handled before sharing?

Honest reporting

A researcher deletes accurate low scores because they weaken the desired conclusion.

What is wrong with this decision?

Selective reporting

A team runs ten comparisons and reports only the one with its preferred result.

What information does the audience need to judge the evidence?

Fabrication and falsification

Fabrication invents observations. Falsification alters or selectively removes observations to misrepresent findings.

What is wrong with copying completed survey responses to fill missing records?

Stopping when results look favorable

A team checks results repeatedly and stops as soon as the preferred conclusion appears.

How could a prespecified data-collection plan improve the reliability of the study?

Graph scales and ethical reporting

Two graphs show the same synthetic rise from 98 to 102. One vertical axis starts at 0; the other starts at 97.

How could the choice of scale influence a reader's impression?

Comparing complaint rates

Synthetic service A has 30 complaints from 1,000 customers; B has 20 from 200.

Would reporting only complaint counts give a fair comparison?